

TISUL', Russian Federation

WMO: 295570

Lat: 55.7517N

Lon: 88.2976E

Elev: 694

StdP: 14.33

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-29.4	-40.7	0.5	-35.6	-33.9	0.8	-29.1	30.1	14.7	26.4	21.8	0.4	170	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	84.3	66.7	80.9	65.3	77.8	63.9	69.4	79.7	67.6	77.4	65.9	75.0	5.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.8	74.4	63.9	91.2	72.6	62.1	85.6	71.0	34.1	79.7	32.5	77.8	31.1	75.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.3	17.7	DB	-39.0	90.4	8.5	2.9	-45.1	92.5	-50.0	94.2	-54.8	95.8	-60.9	98.0	(4)
(5)			WB	-39.4	72.3	7.9	2.9	-45.0	74.4	-49.7	76.0	-54.1	77.6	-59.8	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	1.5	7.4	21.2	37.8	50.6	61.5	64.9	60.7	48.9	36.2	19.8	8.1
	DBStd	24.76	17.28	14.67	12.39	9.95	9.83	7.24	5.17	6.45	8.40	10.14	15.02	15.72
	HDD50	6861	1504	1192	894	378	117	8	0	4	121	437	907	1299
	HDD65	11085	1969	1612	1358	817	454	145	66	165	486	892	1357	1764
	CDD50	1397	0	0	0	12	137	353	462	335	88	10	0	0
	CDD65	145	0	0	0	0	8	40	63	31	3	0	0	0
	CDH74	1623	0	0	0	2	158	511	596	308	48	0	0	0
	CDH80	361	0	0	0	0	39	114	135	63	10	0	0	0
Wind	WSAvg	6.2	6.3	6.7	7.5	7.5	6.6	4.4	3.5	3.8	5.5	7.2	7.4	7.6
Precipitation	PrecAvg	19.9	0.6	0.6	0.7	1.3	1.8	2.9	2.7	3.0	2.2	1.8	1.4	1.1
	PrecMax	24.7	1.2	1.3	2.0	2.6	4.6	5.6	4.5	5.5	5.5	3.4	2.7	2.0
	PrecMin	14.1	0.2	0.1	0.1	0.2	0.5	1.3	1.0	1.0	0.2	0.9	0.6	0.5
	PrecStd	2.6	0.2	0.3	0.4	0.6	0.9	1.0	0.9	1.1	1.1	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	38.9	50.4	70.2	85.3	87.3	88.5	86.2	80.6	66.6	50.9	38.9
		MCWB	33.0	34.0	40.2	52.4	61.1	66.5	69.7	68.3	62.6	50.8	42.6	33.9
	2%	DB	32.8	34.5	44.7	63.9	78.0	83.3	84.1	81.0	72.6	60.6	45.5	34.7
		MCWB	29.4	30.1	37.5	48.9	57.7	65.5	68.3	66.9	58.8	48.6	40.2	31.3
	5%	DB	29.3	31.6	41.1	59.2	73.6	80.0	80.3	77.4	67.7	55.5	41.0	31.5
		MCWB	26.4	28.1	35.3	47.1	55.7	64.2	66.8	64.8	55.8	47.0	36.8	28.5
	10%	DB	25.2	28.3	37.7	54.1	68.8	76.4	77.1	73.6	63.0	50.8	37.5	28.3
		MCWB	22.6	25.3	32.9	44.2	54.1	62.9	65.7	62.6	54.0	44.3	34.1	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.3	34.6	41.6	53.7	63.6	70.4	72.3	71.6	63.8	53.3	44.4	34.9
		MCDB	37.4	38.2	48.2	67.1	81.1	81.8	83.2	81.3	76.2	62.6	48.8	38.3
	2%	WB	29.5	30.7	38.2	50.8	60.2	67.9	70.1	68.2	59.8	50.2	40.2	31.5
		MCDB	32.7	33.9	43.3	62.1	73.6	78.8	80.0	78.7	70.2	58.2	45.1	34.3
	5%	WB	26.1	28.2	35.6	47.6	57.5	66.1	68.3	66.0	57.2	47.4	36.9	28.5
		MCDB	29.1	31.5	40.5	57.8	69.6	77.0	77.6	74.9	65.9	55.0	40.5	31.6
	10%	WB	22.7	25.1	33.1	44.6	55.0	64.1	66.6	63.9	54.8	44.5	34.1	25.3
		MCDB	25.1	28.4	37.6	53.6	67.1	73.9	75.3	72.0	61.9	50.9	37.3	28.2
Mean Daily Temperature Range	5% DB	MDBR	16.6	18.5	18.9	18.5	20.4	20.3	19.2	18.9	18.0	14.2	13.0	15.0
		MCDBR	16.5	15.9	19.6	28.0	29.2	28.1	25.9	25.5	27.7	23.3	15.3	16.6
		MCWBR	14.2	13.2	13.8	16.0	13.4	12.7	11.9	12.6	14.7	13.7	10.9	14.7
	5% WB	MCDBR	15.7	16.0	18.2	25.7	25.4	23.5	21.9	22.8	24.5	22.0	14.1	16.2
		MCWBR	13.6	13.8	13.9	15.9	12.7	12.1	11.3	12.1	13.8	13.5	10.4	14.6
Clear-Sky Solar Irradiance	taub		0.283	0.318	0.367	0.402	0.410	0.405	0.430	0.410	0.370	0.352	0.317	0.285
	taud		2.024	1.998	1.935	2.073	2.194	2.300	2.233	2.265	2.325	2.221	2.046	1.965
	Ebn at Noon		186	226	243	257	263	266	256	253	247	216	168	148
	Edh at Noon		21	32	44	45	42	38	40	37	30	26	21	18
All-Sky Solar Radiation	RadAvg		260	571	1010	1397	1571	1845	1660	1347	889	464	230	160
	RadStd		29	48	78	92	177	172	115	114	108	46	33	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page